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# File: PoojaChaudhari_CyberSecurityProject.py
# IBM SkillsBuild Cyber Security BootCamp 2026
# Project Name: Password Strength Analyzer & Keyboard Auditor Simulator
# Student Name: Pooja Chaudhari

import os
import re
import json
from pynput.keyboard import Key, Listener

print("================================================================")
print("  IBM SkillsBuild Cyber Security Project - BharatCares   ")
print("  Student Name: Pooja Chaudhari                          ")
print("================================================================\n")

def check_password_strength(password):
    """Evaluates password strength based on standard security rules."""
    if len(password) < 8:
        return "WEAK (Password must be at least 8 characters long)"

    if not re.search("[a-z]", password):
        return "WEAK (Missing lowercase letters)"
    if not re.search("[A-Z]", password):
        return "MEDIUM (Missing uppercase letters)"
    if not re.search("[0-9]", password):
        return "MEDIUM (Missing numeric characters)"
    if not re.search("[_@$#!%*?&]", password):
        return "STRONG (Could be excellent with special characters)"

    return "EXCELLENT / VERY STRONG"

def on_press(key):
    """Logs the pressed key to a local audit log file safely."""
    try:
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            with open("security_audit_log.txt", "a") as log_file:
                log_file.write(f"{key.char}")
        except AttributeError:
            with open("security_audit_log.txt", "a") as log_file:
                log_file.write(f" [{key}] ")

def start_keylogger_simulation():
    """Starts the simulation listener for keyboard events."""
    print("[*] Starting Keyboard Audit Simulator...")
    print("[*] Every stroke is being recorded to 'security_audit_log.txt' for analysis.")
    print("[*] Press 'ESC' key to stop the simulator loop safely.\n")

    def on_release(key):
        if key == Key.esc:
            return False

    with Listener(on_press=on_press, on_release=on_release) as listener:
        listener.join()

if __name__ == "__main__":
    # Module 1: Password Strength Test
    print("--- Module 1: Password Security Auditor ---")
    test_pwd = input("Enter a mock password to test strength: ")
    result = check_password_strength(test_pwd)
    print(f"Audit Result: {result}\n")

    # Module 2: Keylogger Simulation
    print("--- Module 2: Endpoint Threat Simulation ---")
    choice = input("Do you want to start the keystroke simulation log? (yes/no): ")
    if choice.lower() == 'yes':
        start_keylogger_simulation()
    print("Project Module executed successfully.")
```